



GROWTH & GH AXIS

MGF

MGF – 2 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Mechano growth factor
SYNONYMS / ALIASES	MGF; IGF-1 Ec (human splice variant); IGF-1Ec C-terminal peptide
CAS NUMBER	[VERIFY WITH COA]
MOLECULAR FORMULA	C121H200N42O39 [VERIFY WITH COA]
MOLECULAR WEIGHT	~2867 g/mol [VERIFY WITH COA]
AMINO ACID SEQUENCE	YQPPSTNKNTKSQRRKGSTFEERK (C-terminal E-domain peptide of the IGF-1Ec splice variant)
SOURCE	Synthetic (solid-phase peptide synthesis)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=98% by HPLC (confirm against batch COA)
CONTAINER	2 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 5-7 days in bacteriostatic water (native MGF is short-lived in solution)
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Avoid repeated freeze-thaw cycles. The unpegylated peptide degrades rapidly in solution; for extended stability see the [[peg-mgf]] variant.

STRUCTURAL NOTES

MGF (mechano growth factor) is a synthetic peptide corresponding to the C-terminal E-domain of the IGF-1Ec splice variant, a mechanically and stress-induced alternative splice product of the insulin-like growth factor-1 gene. The peptide represents the unique C-terminal extension that distinguishes this splice variant from systemic IGF-1; it does not include the mature IGF-1 domain. The pegylated analog is described in the [[peg-mgf]] datasheet.

SELECTED PEER-REVIEWED REFERENCES

1. Yang S. et al. - *Cloning and characterization of an IGF-1 isoform expressed in skeletal muscle subjected to stretch* - Journal of Muscle Research and Cell Motility, 1996 - DOI: 10.1007/BF00124429
2. Hill M., Goldspink G. - *Expression and splicing of the insulin-like growth factor gene in rodent muscle is associated with muscle satellite (stem) cell activation following local tissue damage* - Journal of Physiology, 2003 - DOI: 10.1113/jphysiol.2003.040634

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.